

We proceed with the three steps of the modeling process.

Decision variables The decision variables are the amounts in euros invested for

- state bonds: x_{sb} ,
- real estate loans: x_{re} ,
- car loans: x_{cl} , and,
- scholarship loans: x_{sl} .

Objective function As the company wants to maximize its profit, the objective function must provide the profit as a function of the decision variables:

$$f : \mathbb{R}^4 \rightarrow \mathbb{R} : f(x_{sb}, x_{re}, x_{cl}, x_{sl}).$$

The profit of the investment fund for the following year is calculated based on the interest rates using the following formula:

$$f(x_{sb}, x_{re}, x_{cl}, x_{sl}) = 0.04x_{sb} + 0.06x_{re} + 0.08x_{cl} + 0.09x_{sl}.$$

Therefore, we can write the problem as

$$\max_{x_{sb}, x_{re}, x_{cl}, x_{sl}} 0.04x_{sb} + 0.06x_{re} + 0.08x_{cl} + 0.09x_{sl}.$$

Constraints The total amount to invest is 2.5 M€. This is modeled using the following constraint:

$$x_{sb} + x_{re} + x_{cl} + x_{sl} = 2'500'000$$

Each restriction is modeled as follows:

- the total amount invested in car and scholarship loans ($x_{cl} + x_{sl}$) must not exceed twice the amount invested in bonds ($2x_{sb}$):

$$x_{cl} + x_{sl} \leq 2x_{sb}.$$

- the amount invested in car loans (x_{cl}) must be larger or equal than the amount invested in scholarship loans (x_{sl}):

$$x_{sl} \leq x_{cl}.$$

- the investment in car loans (x_{cl}) should not exceed the investment in real estate loans (x_{re}) by more than 70%:

$$x_{cl} \leq 1.7x_{re}.$$

Finally, we must impose all the decision variables to be non negative:

$$x_{sb}, x_{re}, x_{cl}, x_{sl} \geq 0.$$

Putting everything together, we obtain the following optimization problem:

$$\max 0.04x_{sb} + 0.06x_{re} + 0.08x_{cl} + 0.09x_{sl}$$

subject to

$$\begin{aligned}x_{sb} + x_{re} + x_{cl} + x_{sl} &= 2'500'000, \\x_{cl} + x_{sl} &\leq 2x_{sb}, \\x_{sl} &\leq x_{cl}, \\x_{cl} &\leq 1.7x_{re}, \\x_{sb}, x_{re}, x_{cl}, x_{sl} &\geq 0.\end{aligned}$$

For your information, the optimal solution (rounded to 1/10 of euros) is:

- State bonds: 696'721.3 euros
- Real estate loans: 409'836.1 euros
- Car loans: 696'721.3 euros
- Scholarship loans: 696'721.3 euros

for a total profit of 170'902.6 euros

The constraints are verified:

- Constraint 1: $696'721.3 + 409'836.1 + 696'721.3 + 696'721.3 = 2'500'000.0$,
- Constraint 2: $696'721.3 + 696'721.3 = 1'393'442.62 \cdot 696'721.3 = 1'393'442.6$,
- Constraint 3: $696'721.3 \leq 696'721.3$,
- Constraint 4: $696'721.3 \leq 1.7 \cdot 409'836.1 = 696'721.3$.